Active Directory Home Lab Project Documentation

Project Report: Implementation of a Virtualized Active Directory Environment

1. Executive Summary

This project details the successful deployment of a Basic Active Directory Home Lab designed to simulate a production-grade enterprise network. Utilizing Oracle VirtualBox as the primary virtualization platform, the environment integrates a Windows Server 2019 Domain Controller with essential networking services, including DNS, DHCP, and NAT. Key achievements of this project include the automation of large-scale user provisioning via PowerShell and the successful integration of a Windows 10 Pro client into the centralized domain architecture.

---

2. Technical Architecture and Environment

The lab is built upon a dual-node virtualized architecture designed to balance internal security with external connectivity.

2.1 Virtualization Specifications

The environment was provisioned using the following hardware and software specifications:

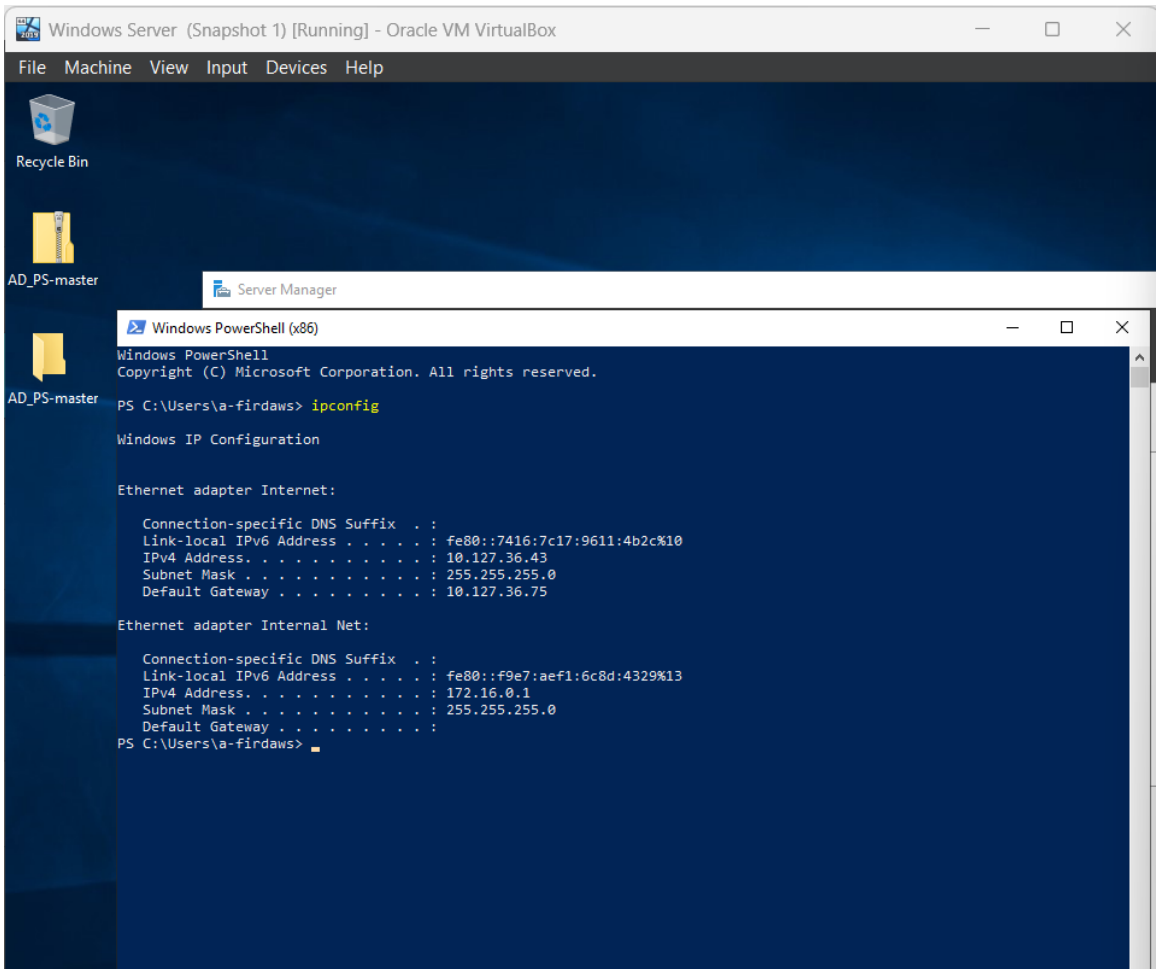
- Hypervisor: Oracle VirtualBox
- Server Infrastructure: Windows Server 2019 (Desktop Experience), allocated with 2GB RAM and 2–4 CPU cores.
- Workstation Infrastructure: Windows 10 Pro (64-bit).

2.2 Network Design

The network is segmented to ensure the Domain Controller (DC) acts as the secure gateway for the internal local area network (LAN).

- External Connectivity: The DC utilizes a NAT adapter for public internet access.

- Internal LAN: A private internal network serves the client workstations. The DC is configured with a static IP address of 172.16.0.1 and a subnet mask of 255.255.255.0.
- Core Services: The DC provides DNS (loopback 127.0.0.1 for local resolution) and manages a DHCP scope ranging from 172.16.0.100 to 172.16.0.200.



---

3. Core Infrastructure Deployment

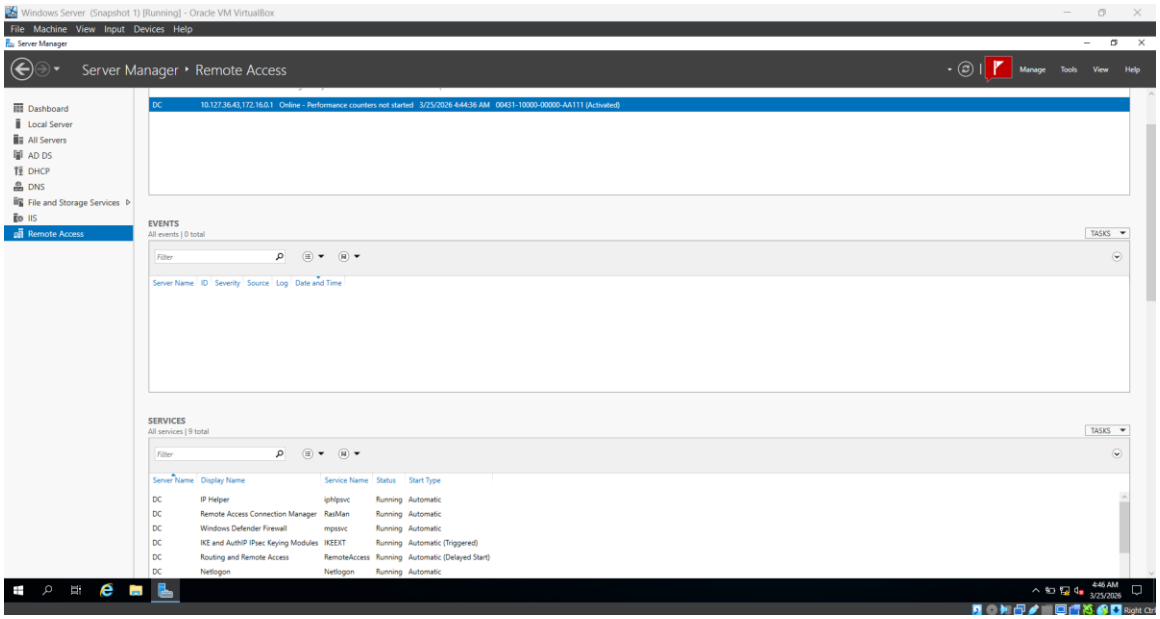
3.1 Active Directory Services

The foundation of the identity management system was established by installing the Active Directory Domain Services (AD DS) role. Following installation, the server was promoted to a Domain Controller, initiating a new forest under the domain mydomain.com. To maintain administrative best practices, a dedicated Organizational Unit (OU) named _admins was created to house a custom Domain Admin account, ensuring that daily operations were not performed using the default built-in administrator.

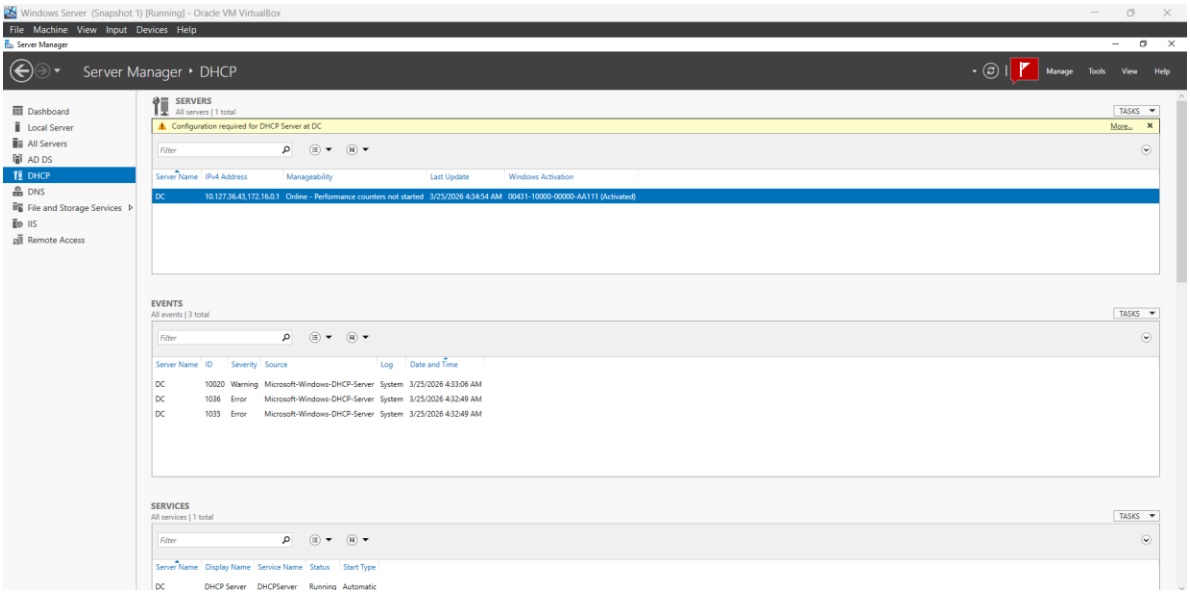
3.2 Network Service Integration

To provide full functionality to the internal network, two critical services were configured on the DC:

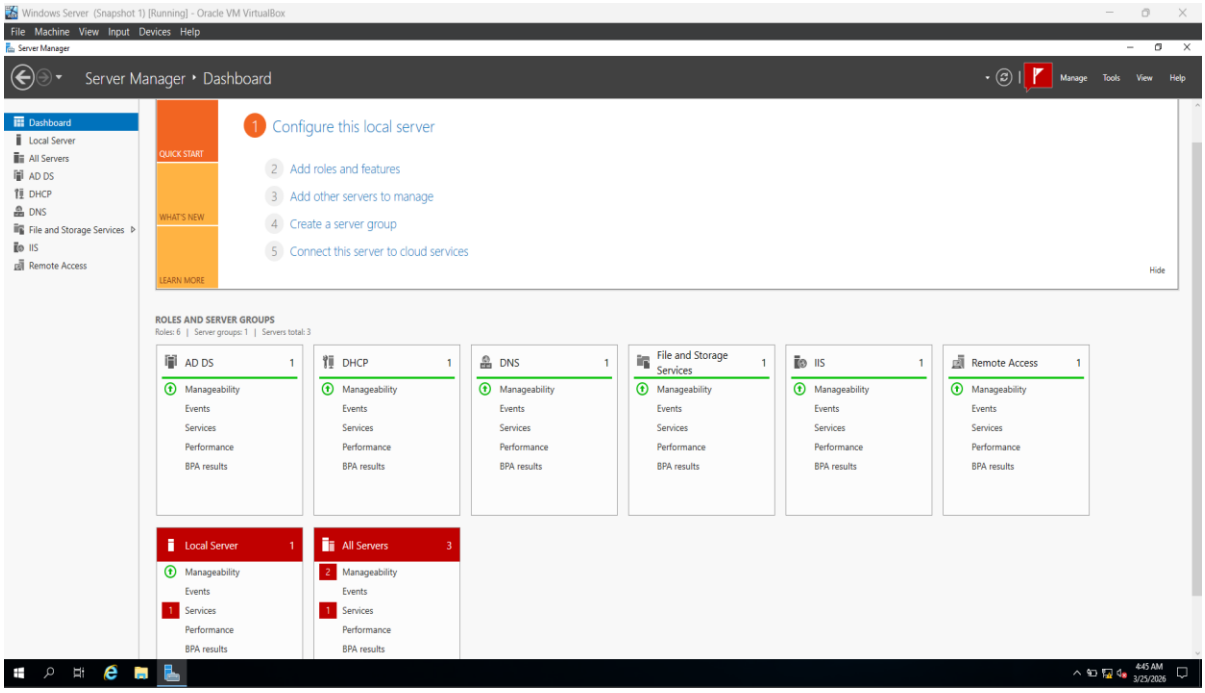
1. Routing and Remote Access (RRAS): Configured to provide Network Address Translation (NAT), allowing internal clients to access the internet through the DC's public interface.



2. DHCP Server: The DHCP role was installed and authorized within Active Directory to automate IP addressing for all client machines, with the DC (172.16.0.1) set as the default gateway.



To provide a more comprehensive view of the automation phase, the Administrative Automation section has been expanded to detail the script's logic, security protocols, and organizational impact based on the project documentation.



---

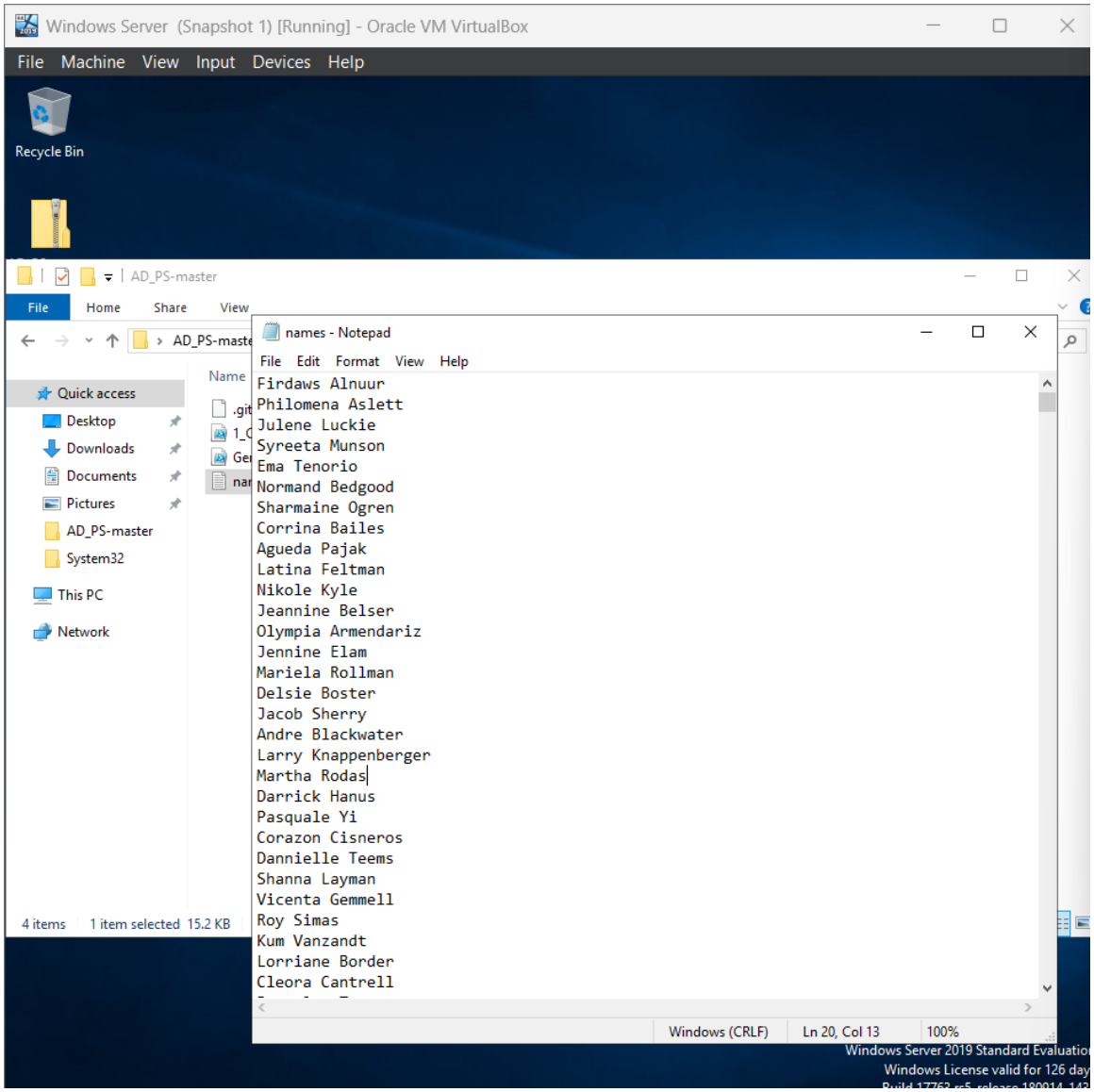
4. Administrative Automation and Bulk Provisioning

To demonstrate administrative efficiency and scalability within the mydomain.com forest, a custom PowerShell script was developed and executed to automate the provisioning of user accounts. This approach highlights the transition from manual entry to repeatable, code-based administration.

4.1 Script Logic and Data Ingestion

The automation process began with the ingestion of identity data. The script was designed to:

- Read from External Data: It processed a text file containing a list of names, allowing for high-volume importing without manual data entry.

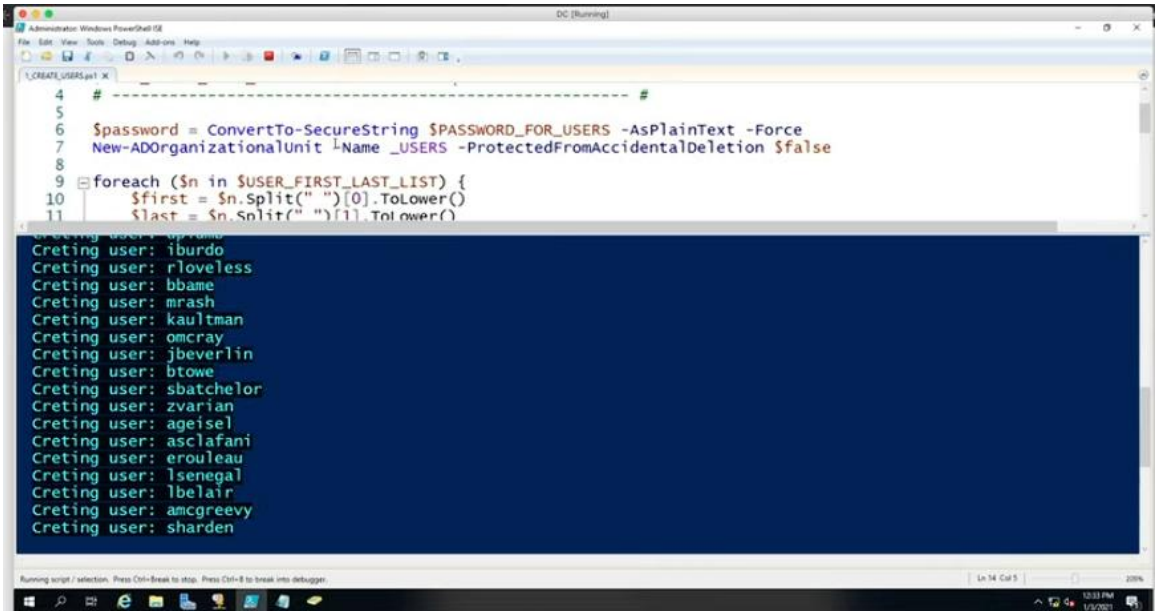


- Automated Naming Conventions: To maintain directory consistency, the script programmatically generated usernames using a standardized "first initial + last name" format (e.g., "falnuur" for firdaws alnuur).

4.2 Directory Structure and Organizational Units

The script did not merely add users to the default container; it managed the directory's logical structure.

- OU Creation: The script included logic to create a dedicated Organizational Unit (OU) named _users.



- Logical Segmentation: By placing the 1,000 automated accounts into this specific OU, the environment remains organized and prepared for future Group Policy Objects (GPOs) targeted specifically at end-users, while keeping them separate from the _admins OU.

4.3 Security and Account Configuration

Security was a primary focus during the automation phase. The script performed several critical functions to ensure the accounts were secure and functional:

- Credential Encryption: Before passing passwords to the Active Directory, the script converted them into secure strings, adhering to the requirement that sensitive data must be encrypted in transit and at rest.
- Account Readiness: Each account was created in an enabled state with specific password policies applied. This ensured that the accounts were immediately ready for use by the Windows 10 client once the deployment was complete.

4.4 Scalability and Verification

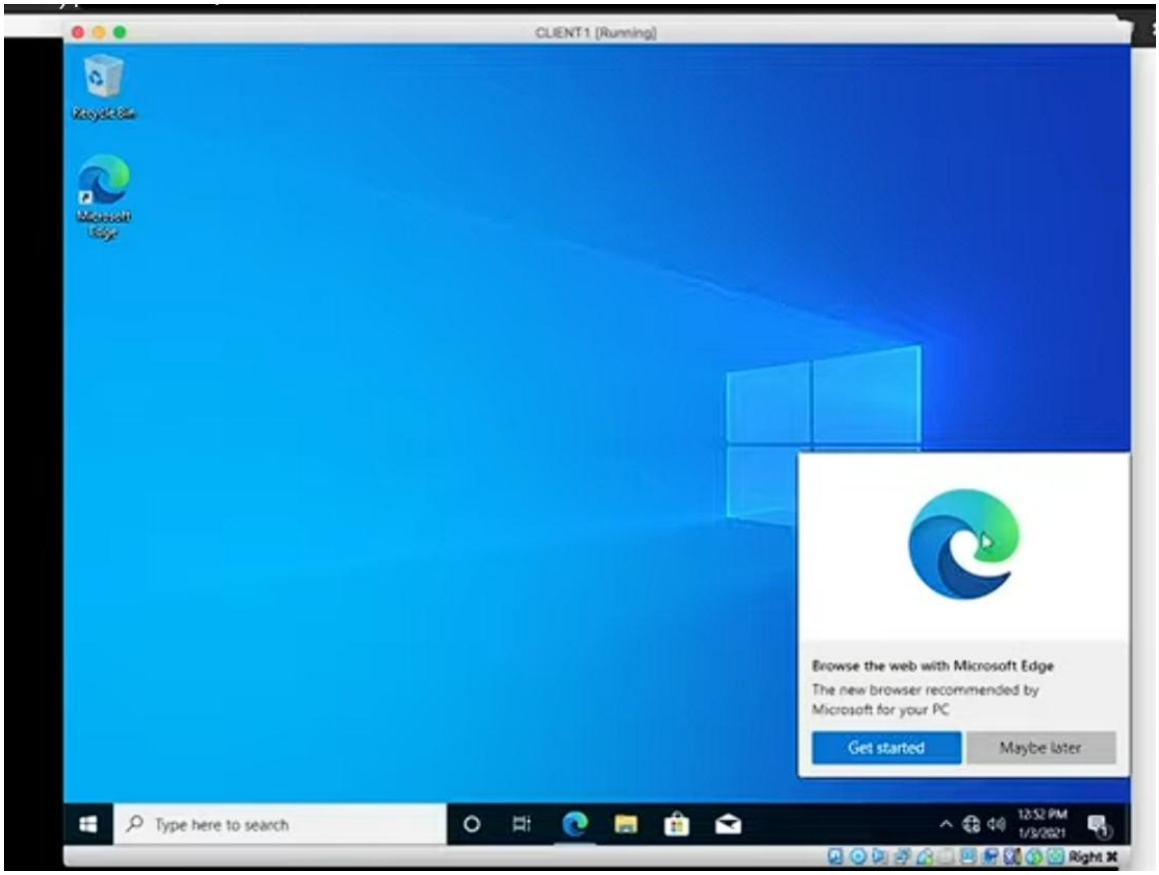
The primary achievement of this automation was the simultaneous creation of 1,000 domain users. This scale serves as a "stress test" for the Domain Controller's ability to handle bulk directory objects. Following execution, the results were verified within the Active Directory Users and Computers (ADUC) console, confirming that the _users OU was correctly populated, and all objects were recognized by the domain.

---

5. Client Integration and Validation

5.1 Domain Join Process

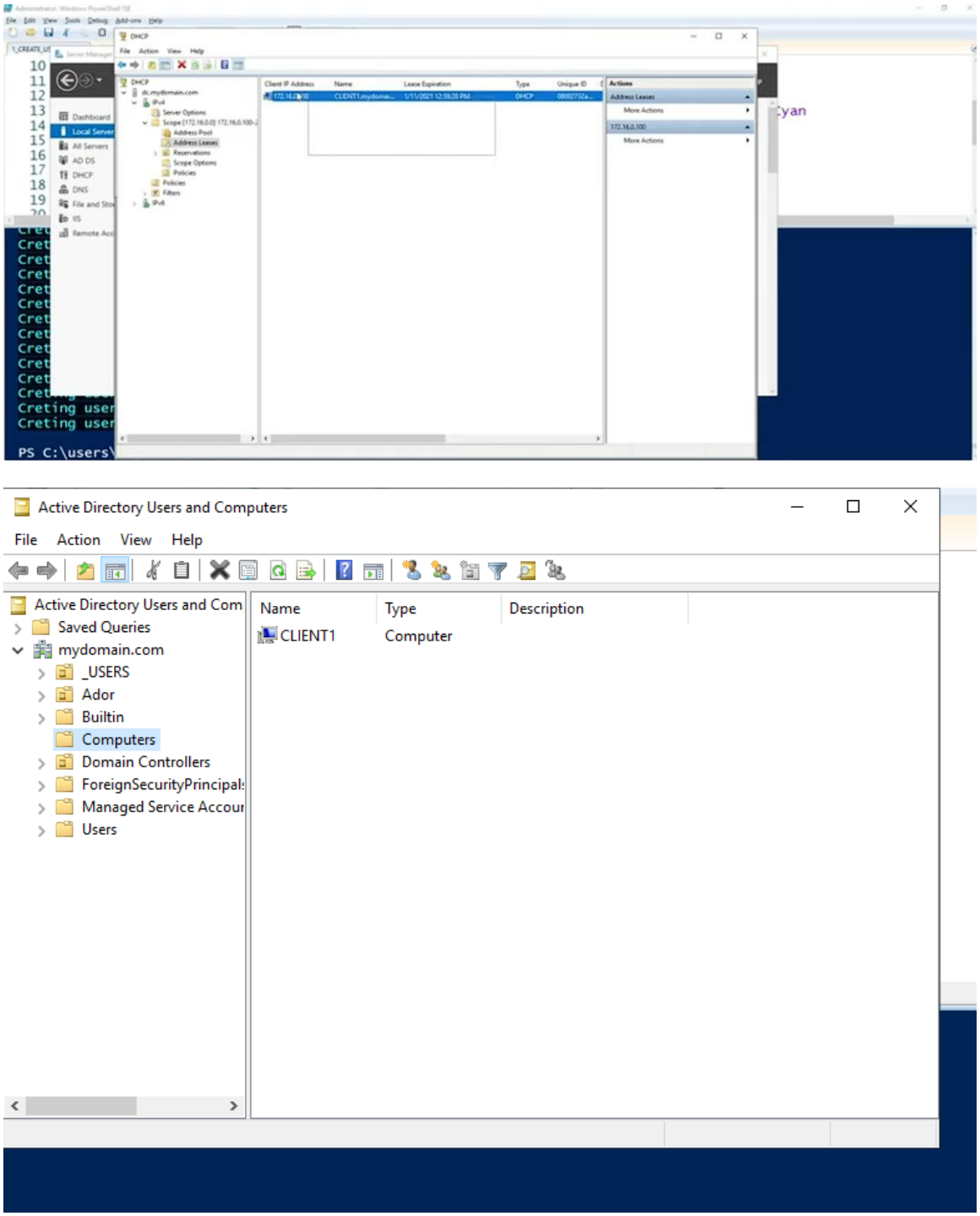
A Windows 10 workstation (identified as client1) was deployed to the internal network. The client successfully obtained an IP lease from the DC's DHCP server and was subsequently joined to the mydomain.com forest. Verification was confirmed by logging into the client using the domain credentials generated during the automation phase.



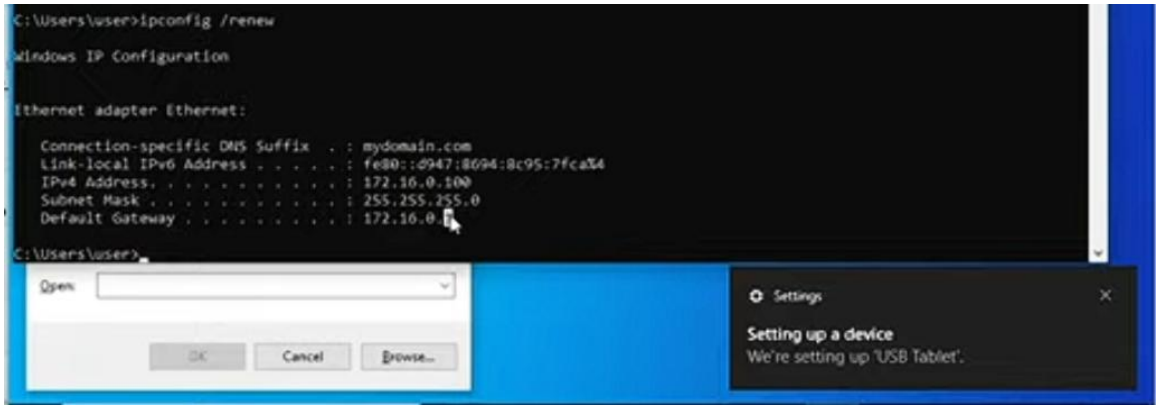
5.2 System Testing and Verification

The integrity of the lab was verified through a series of connectivity and service tests:

- **DHCP Verification:** Confirmed active leases within the DHCP management console.



- **Identity Verification:** Used the whoami command on the client to confirm active domain membership.
- **Connectivity:** Validated that the NAT configuration allowed the client to reach external web resources while remaining on the internal private subnet.



---

6. Conclusion

The implementation of this home lab successfully demonstrates the core components of a Windows-based enterprise environment. By centralizing authentication, automating user management, and providing routed network services, the project establishes a robust foundation for advanced testing in cybersecurity, Group Policy management, and network administration.